Gout

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Gout is a common and treatable disease caused by the deposition of monosodium urate crystals in articular and non-articular structures. Increased concentration of serum urate (hyperuricaemia) is the most important risk factor for the development of gout. Serum urate is regulated by urate transporters in the kidney and gut, particularly GLUT9 (SLC2A9), URAT1 (SLC22A12), and ABCG2. Activation of the NLRP3 inflammasome by monosodium urate crystals with release of IL-1β plays a major role in the initiation of the gout flare; aggregated neutrophil extracellular traps are important in the resolution phase. Although presenting as an intermittent flaring condition, gout is a chronic disease. Long-term urate lowering therapy (eg, allopurinol) leads to the dissolution of monosodium urate crystals, ultimately resulting in the prevention of gout flares and tophi and in improved quality of life. Strategies such as nurse-led care are effective in delivering high-quality gout care and lead to major improvements in patient outcomes.

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See **Online** for appendix

Introduction

Gout is a common condition caused by the deposition of monosodium urate crystals in articular and non-articular structures. A high serum urate concentration is the most important risk factor for the development of gout. In clinical practice and research, hyperuricaemia (blood urate concentration over the saturation threshold) is typically reported when serum urate is higher or equal to 0·42 mmol/L (7 mg/dL). Gout presents as intermittent episodes of severely painful arthritis (gout flares) caused by the innate immune response to deposited monosodium urate crystals. The central strategy for effective management of gout is long-term urate-lowering therapy to reverse hyperuricaemia, which leads to the dissolution of monosodium urate crystals and long-term prevention of gout flares. In this Seminar, we provide an update on the clinical features, pathophysiology, and treatment of gout. Key terms and definitions are provided on appendix p 1.

Clinical presentation of gout

The typical first presentation of gout is an intensely painful acute inflammatory arthritis (gout flare) affecting a lower limb joint.[1] In the absence of treatment, the gout flare is typically self-limiting over a period of 7–14 days. After resolution, there is a pain-free asymptomatic period (intercritical gout), until another gout flare occurs. Over time, some people with persistent hyperuricaemia also develop tophi, chronic gouty arthritis (persistent joint inflammation induced by monosodium urate crystals), and structural joint damage.

Gout flares can occur in the joints or periarticular tissues (eg, bursae, tendons, and entheses). The pain can be described as stabbing, gnawing, burning, or throbbing.[2] Another well described feature of the gout flare pain is its short time from onset to peak intensity (usually less than 12 h).[1] Accompanying features of the flare are varying degrees of swelling, warmth, and erythema.[3] The intensity of the gout flare leads to a substantial limitation in the ability to use the affected area, difficulty walking, and a fear of even minimal physical contact.[1]

The pattern of joint involvement during a gout flare can greatly aid in its recognition. The lower limb (foot, ankle, and knee) is preferentially involved.[1] The involvement of the first metatarsophalangeal joint (podagra) is characteristic.[1,4] Gout flares can also occur in the elbows, wrists, and hand joints, but upper limb involvement usually occurs only in patients with long-standing, poorly controlled disease. Involvement of the axial skeleton occurs occasionally.[5] Gout flares are usually monoarticular, although oligoarticular or polyarticular episodes do occur, typically in patients with poorly controlled disease or during hospitalisation. Polyarticular flares can be associated with pronounced systemic symptoms, including fever, chills, and even delirium.[6]

The recurrence of gout flares is difficult to predict, but the likelihood of recurrent flares is associated with the severity of hyperuricaemia.[7] Triggers for gout flares include meals high in purine, alcohol intake, joint trauma, and acute illness.[8–10] Some patients have a single gout flare episode without any further symptoms, even untreated.[11] However, most patients have recurrent flares.[1] With persistent exposure to increased concentrations of serum urate and increasing monosodium urate crystal burden, each gout flare can last longer and even merge with subsequent flares, minimising or abolishing the duration of the asymptomatic intercritical periods.

Although occasionally presenting early in the course of the disease,[12] tophaceous gout and chronic gouty arthritis are usually manifestations of long-standing disease without adequate serum urate control.[13] Subcutaneous

Search strategy and selection criteria

We searched MEDLINE for publications dated Jan 1, 1946, to Dec 31, 2020, in English language exclusively, using the search term "gout". We largely selected publications from the past 5 years, but did not exclude commonly referenced and highly regarded older publications. We also searched the reference lists of articles identified through this search strategy and selected those we judged relevant. Review articles are cited to provide readers with more details and more references than this Seminar has room for.

tophi present as "draining or chalk-like subcutaneous nodule under transparent skin, often with overlying vascularity, located in typical locations: joints, ears, olecranon bursae, finger pads, tendons (eg, Achilles)".[14] Subcutaneous tophi can vary greatly in size, from barely palpable to large confluent masses (appendix p 2). On palpation, tophi are firm or hard, but softening can occur during resolution with urate-lowering therapy. Although tophi can become acutely inflamed, these lesions are usually not tender, and without warmth or erythema. Depending on location, tophi cause restriction to joint movements, concerns about physical appearance, and difficulty fitting shoes.[15] Joint deformity and associated joint damage are common in patients with tophaceous gout. Ulceration with discharge of a thick, white material (consisting of monosodium urate crystals) and superimposed infection are complications of tophaceous gout.

Gout is commonly associated with other conditions such as hypertension, obesity, cardiovascular disease, diabetes, dyslipidaemia, chronic kidney disease, and kidney stones (appendix p 3).[16–21] These conditions can complicate the management of gout and contribute to premature mortality.[22]

Epidemiology

Incidence and prevalence

Population-based studies from Asia, Europe, and North America have reported incidence ranges between 0·6 and 2·9 per 1000 person-years, and prevalence ranges between 0·68% and 3·90% in adults.[17,18,23–26] The prevalence of gout increased steadily in the 20th century, probably due to the changing age structure of populations and to the growing rates of metabolic syndrome and its associated pathologies. However, the prevalence of gout appears to have stabilised in high-income countries.[24,26]

The prevalence of gout is highly variable between different ethnic groups and across different regions,[27] and no single estimate captures the global prevalence of gout. In the USA, gout is less common in Asian and Hispanic individuals than in African-American and white individuals.[26] Worldwide, the highest prevalence is reported in Oceania, where baseline serum urate concentrations are particularly high among Indigenous populations.[28] Gout prevalence among the Māori (8·5%) and Pacific Peoples living in New Zealand (13·9%) are much higher than in non-Māori, non-Pacific New Zealanders (4·3%).[29] Compared with other New Zealanders, Māori and Pacific Peoples also have an earlier onset of disease by almost a decade.[30]

Gout is more common in men than in women, and in population-based studies from Europe and North America, the male to female sex ratio has been reported to range between 2:1 and 4:1.[24–26] The sex ratio in studies from Asia is much higher, at approximately 8:1.[23] The incidence and prevalence of gout also increase with age, although the typical age of presentation is highly dependent on the population.[23–26] Gout is uncommon in women before menopause,[31] but with increasing age, the male to female sex ratio decreases.[24] Gout in women is associated with a higher comorbidity burden than in men.[32]

Risk factors

Hyperuricaemia is the most important risk factor for the development of gout, and there is a concentration-dependent relationship between serum urate and the risk of incident gout.[33] Many factors that contribute to hyperuricaemia are also risk factors for incident gout, including metabolic syndrome, chronic kidney disease, and medications such as diuretics and ciclosporin. Environmental exposures including purine-rich foods such as red meat and seafood, alcohol (especially purine-rich beer), and sugar-sweetened beverages also increase concentrations of serum urate and the risk of gout.[34–36] In genome-wide association studies, 183 loci that contribute to either high or low serum urate have been identified, 55 of which associate with the risk of gout.[37] The identified genetic loci explain about 7·7% of the variance in serum urate concentrations. The relative effect of genetic and environmental risk factors for gout is controversial. A systematic study of different food groups in a cohort of 16 760 individuals of European ancestry found that, in contrast to genetic contributions, diet explains relatively little of the variance in serum urate concentrations in the general population without gout,[38] but it is unknown whether the same is true for people with gout.

Pathophysiology

The progression of hyperuricaemia and gout can be considered to take place over four pathophysiological stages: development of hyperuricaemia, deposition of monosodium urate crystals, clinical presentation of gout flares due to an acute inflammatory response to deposited crystals, and clinical presentation of advanced disease characterised by tophi.[39] Some patients present with advanced disease without previous gout flares.

Hyperuricaemia

Hyperuricaemia is an essential step in the development of gout (figure 1). Urate is the end-product of purine nucleotide degradation. High-purine diets or other dietary factors (such as alcohol and fructose intake) that induce degradation of purine nucleotides increase serum urate and the risk of incident gout.[34–36] Medical conditions causing high cell turnover, such as psoriasis and myeloproliferative disorders, also lead to increased serum urate concentrations due to overproduction of urate.

Urate excretion is regulated by the kidney and gut, and underexcretion of urate increases serum urate concentrations. The major role of kidney excretion is emphasised by the directly proportional relationship between serum creatinine and urate concentrations.[40] Increased circulating insulin concentrations (in the context of high body-mass index and metabolic syndrome) and diuretics, such as furosemide, also reduce renal excretion of urate.[41,42]

In the kidney, urate is filtered freely by the glomeruli, and excretion is controlled by a group of urate transporters in the proximal tubule (figure 2). On the apical surface, mediating urate transport from the lumen into the cell, are URAT1 (SLC22A12), OAT4 (SLC22A11), OAT10 (SLC22A13), and GLUT9 (SLC2A9).[43–46] Secretory transporters are ABCG2, ABCC4, and NPT1 (SLC17A1).[47–49] The basolateral membrane of the cell has additional transporters also controlling transit of urate, including OAT1 (SLC22A6), OAT2 (SLC22A7), and OAT3 (SLC22A8), which transport urate into the proximal tubule cells (urate secretion),[50,51] and GLUT9, which regulates urate reabsorption.[52]

At present, handling of urate in the kidney is better understood than that in the gut, but similar mechanisms are likely (figure 2). ABCG2 has also been strongly implicated in intestinal urate excretion, with genetic variants in *ABCG2* causing extrarenal urate under-excretion and hyperuricaemia.[53] Other urate transporters might also influence gut urate excretion, but have not been conclusively determined.[54]

The importance of kidney and gut transmembrane transporters has been reinforced through the association of genetic polymorphisms within the genes that encode them and consequently affect urate concentrations.[37,55] Variants in *SLC2A9* (encoding GLUT9), *ABCG2*, and *SLC22A12* (encoding URAT1) have the strongest association with both serum urate and gout in multiple populations worldwide.[37]

Monosodium urate crystallisation

Although hyperuricaemia is present in virtually everyone with gout, it is not sufficient for the development of clinically apparent disease because most people with hyperuricaemia are asymptomatic and do not develop gout.[33] Even in people with severe hyperuricaemia (higher or equal to 0·60 mmol/L; 10 mg/dL), less than half develop gout over a period of 15 years.[56] Monosodium urate crystal deposition in people with hyperuricaemia is considered the next checkpoint for progression to clinically evident gout; approximately 25% of people with hyperuricaemia have monosodium urate crystal deposition that can be detected through specialised imaging methods, particularly at the first metatarsophalangeal joint.[57,58] The concentration of urate is the most important contributor to the formation of monosodium urate crystals, but low temperatures, physiological pH between 7 and 10, high concentration of sodium ions, and synovial and cartilage components also promote monosodium urate crystallisation.[59–61]

The acute inflammatory response to monosodium urate crystals

Crystalline monosodium urate is a damage-associated molecule and can stimulate innate immune pathways.

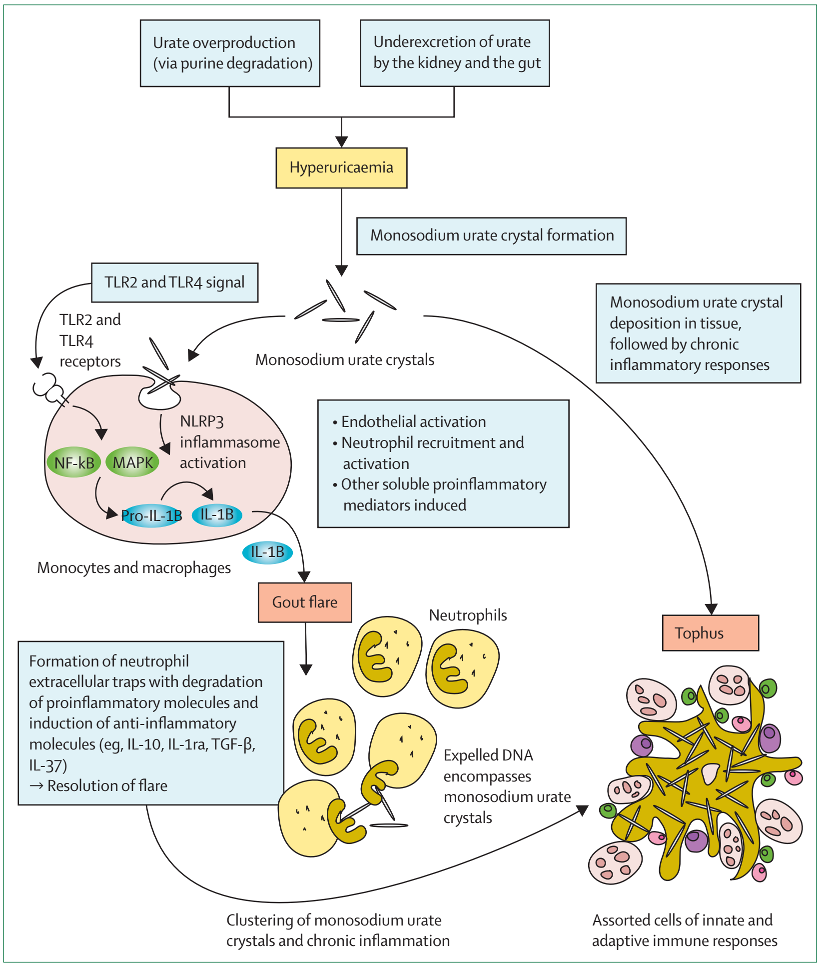

Figure 1: **Progression of hyperuricaemia to gout**
Urate overproduction or underexcretion leads to the development of hyperuricaemia, a state that facilitates the formation of monosodium urate crystals. These crystals can be phagocytosed by monocytes; an additional signal, through TLR2 and TLR4, is required to complete activation of the NLRP3 inflammasome and the production of proinflammatory IL-1β, which leads to acute flares of gouty arthritis. Flare resolution involves neutrophil extracellular traps that bind monosodium urate crystals. Aggregated neutrophil extracellular traps might also contribute to the formation of tophi.

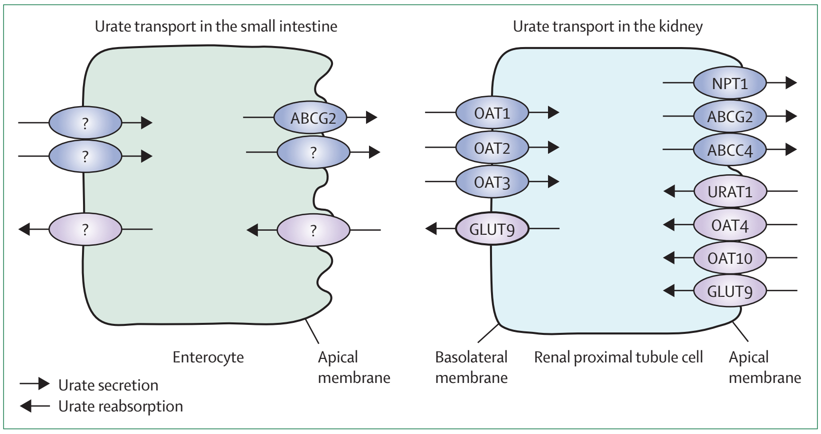

Figure 2: **Location of urate transmembrane transporters in the small intestine and renal proximal tubule cells**

Activation of the NLRP3 inflammasome in macrophages and monocytes by monosodium urate crystals is of particular relevance for the initiation of the gout flare.[62] The NLRP3 inflammasome is dependent on a two-signal initiation system, which avoids unregulated activation that would damage the host. The first signal results in stimulation of NF-κB via TLR4 and TLR2, and synthesis of pro-IL-1β and inflammasome components.[63,64] Monosodium urate crystals act as the second activation signal, causing the assembly of the inflammasome and activation of caspase-1, which proteolyses pro-IL-1β to mature IL-1β.[65] IL-1β then interacts with the IL-1β receptor to trigger a downstream signalling cascade involving proinflammatory cytokines and chemokines, resulting in the recruitment of neutrophils and other cells to the site of crystal deposition.

Because the initiation of the NLRP3 inflammasome requires a two-step initiation process, the deposition of monosodium urate crystals alone does not necessarily result in inflammation, explaining how monosodium urate crystals can be present in the joint without clinically apparent inflammation.[66,67] Factors that stimulate the first signal, such as free fatty acids (induced by intake of a large meal or alcohol), gut microbiota, and other microbiological components, can induce the acute inflammatory response in the presence of deposited monosodium urate crystals.[68–70] AMPK (PRKAA2), a metabolic biosensor with anti-inflammatory activity, inhibits these activation pathways.[71]

Neutrophils produce an array of different proinflammatory substances and mechanisms that allow them to drive local acute immune reactions. However, they can also shut these inflammatory processes down rapidly by ejecting their DNA into the extracellular matrix, known as the deployment of neutrophil extracellular traps. At high neutrophil densities, aggregated neutrophil extracellular traps capture and degrade proinflammatory mediators.[72] This response, together with other anti-inflammatory factors such as IL-1ra (IL1RN), IL-10, TGF-β, and IL-37 contribute to the resolution of the gout flare, despite the persistence of monosodium urate crystals at the site of inflammation.[73,74]

Advanced gout

Advanced gout is characterised by tophi, chronic gouty synovitis, and structural joint damage. The tophus represents a chronic, foreign-body granulomatous inflammatory response to monosodium urate crystals, and consists of three main zones: a core of tightly packed monosodium urate crystals, a surrounding cellular corona zone, and an outer fibrovascular zone.[75] Cells of both the innate and adaptive immune system are present within the tophus, including multinucleated giant cells.[76] Structural joint damage, with bone erosion and focal cartilage damage, is a common feature of advanced gout. Tophi are usually present at sites of structural damage in gout,[77] with numerous bone-resorbing osteoclasts at the tophus-bone interface.[78] In addition, monosodium urate crystals directly reduce osteoblast viability and function,[79] and indirectly promote a shift in osteocyte function, favouring bone resorption and inflammation.[80]

Differential diagnosis

The most important differential diagnosis in the acute clinical setting is soft tissue or musculoskeletal infection (eg, cellulitis, septic bursitis, septic arthritis, or osteomyelitis). Similarly to gout, soft tissue or musculoskeletal infections affect individuals more frequently as they age, or if they have multiple comorbidities, or are transplant recipients. Fever and leukocytosis can be reported in both severe gout flare and infection. Importantly, gout and infection can coexist. Sampling of synovial fluid or other involved tissue for cell counts and Gram-positive bacteria stain and culture should be pursued to fully exclude infection. Imaging techniques can also assist in differentiating gout from bone and joint infections: the presence of high-grade bone oedema on MRI, for example, is highly suggestive of osteomyelitis in people with gout.[81]

Other forms of crystal arthritis such as acute calcium pyrophosphate crystal arthritis (so-called pseudogout) or basic calcium phosphate crystal periarthritis can mimic the gout flare. In the case of calcium pyrophosphate deposition (in which the wrist and knee are the most often affected), joint involvement patterns can help to differentiate crystal arthritis from gout. Chondrocalcinosis on imaging tests is also suggestive of calcium pyrophosphate deposition, but microscopic confirmation of calcium pyrophosphate crystals is the most reliable method of differentiation. Occasionally, both calcium pyrophosphate and monosodium urate crystals are present and detectable by synovial fluid microscopy. Acute basic calcium phosphate crystal periarthritis is less likely to be confused with the gout flare given its preference for the shoulder joints, which are uncommonly involved in gout.

There are various similarities between the clinical presentation of gout and psoriatic arthritis. Patients with psoriasis and psoriatic arthritis have a high prevalence of metabolic syndrome and hyperuricaemia.[82] Similarly to gout, psoriatic arthritis presents as a monoarthritis or oligoarthritis. Involvement of the toes, common in psoriatic arthritis, can be sometimes hard to differentiate from gout. The coexistence of both conditions is also well documented.[83]

Advanced polyarticular gout can mimic the symmetric, small-joint polyarthritis of rheumatoid disease. In addition, rheumatoid nodules can be confused with, or hard to differentiate from, subcutaneous tophi. Monosodium urate crystal deposition is more common in joints affected by osteoarthritis[84] and, occasionally, osteoarthritis mimics tophaceous gout, particularly in the first metatarsophalangeal joint or the hand interphalangeal joints.

Clinical investigations

Microscopic confirmation of monosodium urate crystals in synovial fluid or tophi is considered the gold standard for gout diagnosis. Monosodium urate crystals appear needle-shaped and negatively birefringent under polarising light microscopy (figure 3). In synovial fluid, crystals range in length from 1 μm to 20 μm, but can measure up to 40 μm in tophi.[85]

Gout can also be diagnosed with a high level of certainty and without recourse to joint aspiration in the presence of typical signs and symptoms (eg, podagra).[86] A diagnostic compound score for gout has been developed and validated for patients presenting with monoarthritis in primary care settings. This score comprises the following items: male sex, previous patient-reported arthritis attack, onset within 1 day, joint redness, first metatarsophalangeal joint involvement, hypertension or cardiovascular disease, and high concentration of serum urate.[86] A score of less than 4 ruled out gout in more than 97% of patients, and more than 80% of patients with a score of 8 or higher had gout.[86] The diagnostic rule relies heavily on the presence of hyperuricaemia. Importantly, serum urate concentration is often normal during a gout flare due to increased renal excretion during the acute phase response, and it should be rechecked 2–4 weeks after flare resolution.[87,88] Additionally, hyperuricaemia affects more than 20% of men and 4% of women in the general population,[26] and is not sufficient to make a diagnosis of gout in the absence of symptoms.

In the event of diagnostic uncertainty and if joint aspiration is not available or not feasible, ultrasonography and dual energy CT (figure 4A, B) can aid in the diagnosis of gout.[89,90] The ultrasonographic findings of gout include a double contour sign (reflecting monosodium urate crystal deposition on the surface of hyaline articular cartilage), intra-articular or intrabursal tophi, and a snowstorm appearance. Importantly, the absence of monosodium urate crystal deposits on ultrasonography or dual-energy CT does not exclude gout, particularly early in the disease. Plain radiographs are usually normal at the time of initial diagnosis. In late-stage disease, characteristic features include defined, juxta-articular punched out erosions with sclerotic margins and overhanging edges (figure 4C). Joint space width is generally preserved except in advanced tophaceous gout, at which time joint space widening or narrowing can occur (figure 4C).

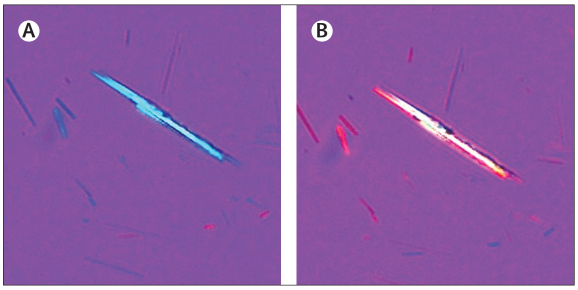

Figure 3: **Polarising light microscopy of synovial fluid demonstrating negatively birefringent monosodium urate crystals**
Left panel perpendicular to lambda axis, and right panel parallel to lambda axis, both at 40× magnification.

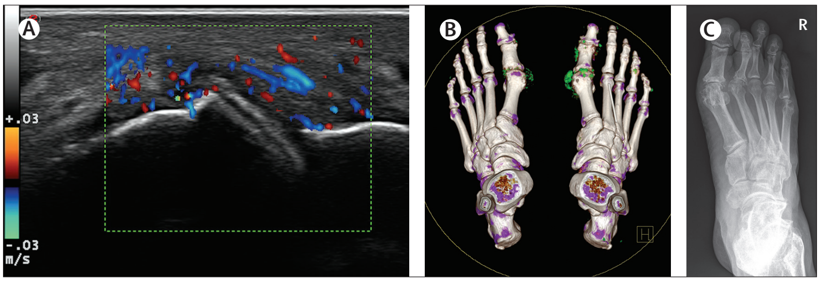

Figure 4: **Radiology of gout**
(A) Ultrasonography scan of the dorsal aspect of the first metatarsophalangeal joint showing the double contour sign, which reflects monosodium urate crystal deposition on the surface of the articular hyaline cartilage; image acquired during a flare, showing intense colour doppler signal. Image courtesy of Philip Courtney. (B) Dual-energy CT of a patient with gout, showing deposition of monosodium urate crystals (shown in green), particularly at the first metatarsophalangeal joints. (C) Plain radiograph of the right foot demonstrating erosive gout in a patient with extensive subcutaneous tophi. Note the involvement of the first, second, and fifth metatarsophalangeal joints, and toes.

Management of the gout flare

The major priorities in the management of the gout flare are pain control and suppression of joint inflammation. Early administration of anti-inflammatory treatment (table 1) is recommended to rapidly suppress joint pain and inflammation. Head-to-head clinical trials comparing oral agents with different mechanisms of action have shown equivalent efficacy between oral prednisolone, non-steroidal anti-inflammatory drugs (non-selective or COX-2-selective), and low-dose colchicine for gout flare management.[91–94] These studies have indicated that oral corticosteroids might have a slightly better safety profile than non-steroidal anti-inflammatory drugs,[92,93] and that fewer side-effects are caused by non-steroidal anti-inflammatory drugs than by low-dose colchicine.[91] Low-dose colchicine (1·0–1·2 mg immediately, followed by 0·5–0·6 mg after 1 h) leads to fewer side-effects compared with high-dose colchicine (4·8 mg administered over 6 h) with similar efficacy,[95] and high-dose colchicine is not recommended.[96,97]

For patients who have had recurrent gout flares, the choice of agent is often determined by patient preference based on previous experience of efficacy and adverse events. For many patients, comorbidities and concomitant medications also influence the choice of agent. For example, non-steroidal anti-inflammatory drugs are avoided if there is concomitant kidney disease, cardiac disease, peptic ulcer disease, and anticoagulant use. High-dose oral prednisone is avoided in the setting of infection, fluid overload, or diabetes. Colchicine toxicity can occur in patients with severe kidney disease, severe liver disease, and concomitant use of strong P-glycoprotein 1 (ABCB1) inhibitors, CYP3A4 inhibitors, or both (eg, ciclosporin, ketoconazole, clarithromycin,

	Key points	Examples of intervention
Anti-inflammatory pharmacological therapy	Early administration of anti-inflammatory therapy is recommended; first-line therapy can be oral corticosteroids, non-steroidal anti-inflammatory drugs, or colchicine; the choice of first-line oral agent is dependent on individual patient factors, including comorbidities, concomitant medications, and previous experience of efficacy and side-effects; IL-1 inhibitors are reserved for patients who are unable to tolerate or have contraindications to the first-line anti-inflammatory therapy	Non-steroidal anti-inflammatory drugs (naproxen 500 mg twice daily orally for 5 days or etoricoxib 120 mg daily orally for 8 days); colchicine (1·2 mg orally at the time of the gout flare, then 0·6 mg orally after 1 h [total dose 1·8 mg]); corticosteroids (prednisolone 30 mg orally daily for 5 days or intra-articular triamcinolone acetonide 40 mg into inflamed knee); IL-1 inhibitors (anakinra 100 mg subcutaneously daily for 3–5 days or canakinumab 150 mg subcutaneously as a single injection)
Non-pharmacological therapy for gout flare	Adjuvant to anti-inflammatory pharmacological therapy	Application of ice pack on the inflamed joint for 30 min four times per day
Assess and support patient disease understanding	Patient education should focus on the concept of gout as a chronic disease caused by monosodium urate crystal deposition, and the potential to dissolve monosodium urate crystals and improve clinical outcomes with long-term urate-lowering therapy	Assess patient's beliefs and priorities; short educational intervention about the cause of gout, the potential to address the underlying basis of the disease, and how to prevent future gout flares and joint damage with long-term urate-lowering therapy
Review and implement long-term management plan	The gout flare presentation is an opportunity to optimise long-term gout management; the severity of pain at the time of the gout flare presentation might be a barrier to full understanding of the long-term management plan; if this is the case, a follow-up should be arranged shortly after the acute presentation to ensure that the patient can discuss the long-term treatment plan in a non-acute setting	Serum urate testing and ambulatory clinical review after resolution of the gout flare to discuss long-term management; written, personalised instructions should be provided by the clinician

Table 1: **Gout flare management strategies (medications vary between countries according to availability and approvals)**

and verapamil).[98] For a single inflamed joint, intra-articular corticosteroids might be the preferred strategy.[99] Intra-articular, intramuscular, or intravenous corticosteroids are also an option for patients experiencing a gout flare who are unable to take oral medications.

Consistent with the central role of the NLRP3 inflammasome activation in the acute inflammatory response to monosodium urate crystals, IL-1 inhibitors are also effective for gout flare management. A single subcutaneous injection of canakinumab 150 mg provided better pain and inflammation relief and reduced the risk of new flares than did treatment with intramuscular triamcinolone acetonide 40 mg, but caused higher rates of adverse events, including serious infections.[100] Anakinra 100 mg subcutaneously daily for 5 days has an efficacy and safety profile similar to first-line oral anti-inflammatory therapies.[101] IL-1 inhibitors are generally reserved for patients who have intolerable side-effects or have contra-indications to first-line anti-inflammatory therapy.

Non-pharmacological therapies (eg, application of ice packs on the inflamed joints) can provide some analgesia.[102] Supportive care including rest, mobility assistance, and adequate nutrition and hydration are important for patients experiencing severe joint pain.

Although the focus of the gout flare consultation is acute symptom management, this interaction represents an important opportunity to ensure patient understanding about gout and optimise long-term gout management. All patients presenting with a gout flare should be informed about the availability of long-term, urate-lowering therapy to address the underlying cause of the disease and prevent future flares and joint damage.[97]

Long-term management of gout

Principles of management

Table 2 outlines the principles of long-term management, with specific examples of treatment. The central strategy for effective, long-term gout management is continuous urate-lowering therapy prescribed at a dose that achieves monosodium urate crystal dissolution, using a treat-to-serum urate target approach.[97,103] For most people with gout, the target serum urate is under 0·36 mmol/L (6 mg/dL), but for those with high urate burden (such as tophaceous gout), a lower serum urate target, of 0·30 mmol/L (5 mg/dL) or less might be needed.[97] There is now evidence to support the treat-to-serum urate target approach, with clinical trials showing that, in the long-term, this treatment strategy leads to suppression of gout flares, regression of tophi, and prevents joint damage.[104–106]

All patients should be informed that gout is a chronic disease of monosodium urate crystal deposition, and that long-term urate-lowering therapy can dissolve monosodium urate crystals and improve clinical outcomes. The 2016 European League Against Rheumatism gout management guidelines recommend urate-lowering therapy for "all patients with recurrent flare (≥2/year), tophi, urate arthropathy and/or renal stones".[97] In addition, "initiation of ULT [urate-lowering therapy] is recommended close to the time of first diagnosis in patients presenting at a young age (<40 years), or with a very high SUA [serum uric acid] level (>8 mg/dL; 480 µmol/L) and/or comorbidities (renal impairment, hypertension, ischaemic heart disease, heart failure)".[97] Similar recommendations for urate-lowering therapy have been made by the American College of Rheumatology.[103]

	Key points	Examples of intervention
Assess and support patient disease understanding	Patient education should focus on the concept of gout as a chronic disease of monosodium urate crystal deposition, and the potential to dissolve monosodium urate crystals and improve clinical outcomes with long-term urate-lowering therapy; delivery of this intervention by nurses leads to improved clinical outcomes	Assess patient's beliefs and priorities; comprehensive educational intervention about the cause of gout, and the potential to address the underlying basis of disease and prevent future gout flares and joint damage with long-term urate-lowering therapy; written, personalised instructions should be provided by the clinician
Assess whether urate-lowering therapy is indicated and, if indicated, start allopurinol as a first-line urate-lowering therapy	Indications for urate-lowering therapy are any of the following in a person with gout: gouty tophi, radiographic damage due to gout, frequent gout flares (two or more flares over a 1 year period), and urolithiasis; also recommended in patients presenting at a young age (<40 years), with a very high serum uric acid concentration (>0·48 mmol/L, 8 mg/dL), or comorbidities (kidney or heart disease); for most people with gout, allopurinol is the first-line urate-lowering therapy and can be started during a gout flare; starting urate-lowering therapy at a low dose with gradual dose escalation can reduce the risk of gout flares	If indicated, prescribe allopurinol 100 mg daily; if estimated glomerular filtration rate is <60 mL/min per 1·73 m², start allopurinol at a lower dose (50 mg daily)
Prevent future gout flares with anti-inflammatory prophylaxis	Subsequent gout flares can be prevented with low-dose anti-inflammatory agents; anti-inflammatory prophylaxis is recommended for 3–6 months when initiating urate-lowering therapy	Naproxen 250 mg daily for 3 months; colchicine 0·6 mg daily for 3 months
Action plan for future gout flares	So-called pill in the pocket medication that allows the patient to treat the gout flare as soon as possible	Naproxen 500 mg orally twice daily for 5 days; etoricoxib 120 mg orally daily for 8 days; colchicine 1·2 mg orally at the time of the gout flare, then 0·6 mg after 1 h (total dose 1·8 mg); prednisolone 30 mg orally daily for 5 days
Treat-to-target care	The serum urate target should be maintained long term to achieve monosodium urate crystal dissolution, prevention of gout flares, progression of tophi, and prevention of joint damage; urate-lowering therapy should be adjusted to achieve and maintain the serum urate target; once the serum urate target is achieved, the serum urate should be monitored periodically (every 6–12 months) to ensure that this target is maintained in the long term	Serum urate target: <0·36 mmol/L (6 mg/dL) for most patients on urate-lowering therapy; lower target (eg, <0·30 mmol/L, 5 mg/dL) for patients with large monosodium urate crystal burden
Optimise urate-lowering therapy	Allopurinol should be increased to achieve the serum urate target; if allopurinol is not tolerated or target serum urate cannot be reached with allopurinol monotherapy, consider other oral urate-lowering therapies, choice dependent on patient factors and drug availability; if oral urate-lowering therapy does not result in serum urate target, and there are frequent gout flares or tophi, switch to pegloticase	Allopurinol monotherapy: allopurinol dose escalation by 100 mg every month until serum urate target achieved (50 mg increments if estimated glomerular filtration rate <60 mL/min per 1·73 m²), maximum 800–900 mg daily; examples of other urate-lowering therapies: febuxostat 40-120 mg daily; probenecid up to 1 g twice daily as monotherapy or in combination with allopurinol; benzbromarone 50–200 mg daily as monotherapy or in combination with allopurinol; pegloticase 8 mg intravenous infusion every 2 weeks
Address questions about diet	Modest reductions in serum urate can be achieved with weight loss and diet modification, such as the Dietary Approaches to Stop Hypertension diet, but most people with gout will not achieve serum urate target with dietary management alone; patients might identify specific foods that trigger gout flares, and avoidance of these dietary triggers is recommended while establishing urate-lowering therapy; avoidance of food triggers is not necessary once urate-lowering therapy is established and monosodium urate crystal dissolution is achieved	Support weight loss if patient is overweight or obese; recommend avoidance of sugar-sweetened drinks; recommend reduction of alcohol in case of hazardous drinking; recommend avoidance of specific foods that patient identifies as triggers while establishing urate-lowering therapy
Manage associated medical conditions	Gout is commonly associated with hypertension, obesity, cardiovascular disease, diabetes, dyslipidaemia, chronic kidney disease, and kidney stones; these conditions can complicate the management of gout and contribute to premature mortality in people with gout	Assess for associated medical conditions (eg, measure body-mass index, blood pressure, creatinine, and glycated haemoglobin); actively manage cardiovascular risk; consider medications that can have benefits for both urate-lowering and related medical conditions (eg, losartan and SGLT2 [SLC5A2] inhibitors)
Ensure ongoing urate-lowering therapy supply and follow-up	Plan long-term follow-up to ensure that disease understanding is reinforced and that the patient has a long-term supply of urate-lowering therapy	Regular serum urate testing and prescription renewal after serum urate target is achieved

Table 2: **Principles of long-term gout management (medications vary between countries according to availability and approvals)**

Urate-lowering therapy can be started during a gout flare,[107,108] and should be started at a low dose with gradual dose titration to reduce the risk of flares that commonly occur in the first few months of initiating urate-lowering therapy.[109] Gout flares can also be prevented during this period with low-dose anti-inflammatory medications such as naproxen 250 mg daily or colchicine 0·6 mg daily for 3–6 months.[109–111] All patients should have an action plan for future gout flares, ideally with a so-called pill in the pocket medication that allows the patient to treat the flare as soon as possible.

Patients often have questions about dietary management of gout.[112] Modest reductions in serum urate can be achieved with diet modification, such as the Dietary Approaches to Stop Hypertension diet and weight loss.[113,114] Sugar-sweetened drinks and excessive alcohol intake should be avoided. However, most people with gout do not reach serum urate target with dietary management alone.[115] Patients can identify specific foods that trigger gout flares, and avoidance of these dietary triggers might be of benefit.[8,116] Avoidance of food triggers is not usually necessary once urate-lowering therapy is established and monosodium urate crystal dissolution is achieved.[117]

Assessment for and management of associated medical conditions (eg, measuring body-mass index, blood pressure, creatinine, and glycated haemoglobin) is recommended.[97] Some medications used to treat comorbid conditions, such as losartan for hypertension, sodium-glucose co-transporter inhibitors for diabetes, and fenofibrate for dyslipidaemia, have modest urate-lowering properties.[118,119]

Follow-up care should include processes to ensure that disease understanding is reinforced, and that the patient has an ongoing supply of urate-lowering therapy. A comprehensive educational intervention that includes discussion about the patient's beliefs about gout and treatment priorities leads to high adherence to urate-lowering therapy, achievement of serum urate targets, and improved clinical outcomes.[106] Intermittent serum urate monitoring provides an opportunity to assess progress and treatment adherence, and to identify patients who require additional support or therapy to maintain effective long-term disease control.

Serum urate-lowering medications

Detailed information about urate-lowering medications is provided on the appendix (p 6). The first-line medication for most people with gout requiring urate-lowering therapy is allopurinol,[103] a purine-based xanthine oxidase inhibitor that inhibits the production of urate. When dose-escalated to the serum urate target, most patients with gout can achieve the serum urate target on allopurinol monotherapy.[120,121] Allopurinol is generally well tolerated, but a rash requiring cessation of therapy occurs in 1–2% of patients. In addition, allopurinol can cause the rare but potentially life-threatening allopurinol hypersensitivity syndrome, typically occurring in the first few months of allopurinol therapy, characterised by severe cutaneous reactions (Stevens-Johnson syndrome or toxic epidermal necrolysis), acute kidney injury, hepatitis, and eosinophilia.[122] Allopurinol hypersensitivity syndrome is most common in people of southeast Asian and African-American ethnicity,[123] and is strongly associated with *HLA-B*58:01*.[124] In populations with high rates of *HLA-B*58:01* (eg, up to 20% of Han Chinese living in Taiwan), screening for the risk allele and avoidance of allopurinol leads to substantial reduction in the incidence of allopurinol hypersensitivity syndrome.[125] Additional risk factors for allopurinol hypersensitivity syndrome are chronic kidney disease, diuretic use, and high starting allopurinol dose.[126] Low starting doses, such as 100 mg daily for patients with normal kidney function and 50 mg daily for patients with estimated glomerular filtration rate under 60 mL/min per 1·73 m², are recommended for all patients starting allopurinol.[2,97,103] For patients who tolerate starting allopurinol, the dose can be safely increased by 100 mg increments every month (50 mg increments if estimated glomerular filtration rate is under 60 mL/min per 1·73 m²) until the serum urate target is reached.[120] The maximum dose of allopurinol approved by the US Food and Drug Administration (FDA) is 800 mg daily; outside the USA, the maximum approved dose is 900 mg daily. The average allopurinol dose required to reach serum urate concentrations under 0·36 mmol/L is approximately 400 mg daily.[120] Factors that affect the dose of allopurinol required to achieve the serum urate target are weight, diuretic use, kidney function, and *ABCG2* genotype.[127,128]

Febuxostat is a potent non-purine xanthine oxidase inhibitor.[129] This medication is used as a second-line urate-lowering therapy, at a dose of 40–120 mg daily (the maximum dose approved in the USA is 80 mg daily). Hypersensitivity reactions and deranged liver function tests have been reported with febuxostat, and, as with allopurinol, prescribing of febuxostat with azathioprine or mercaptopurine should be avoided. An FDA-mandated postmarketing trial (CARES) compared the cardiovascular safety of febuxostat (up to 80 mg daily) with allopurinol (up to 600 mg daily) in 6190 patients with gout and major cardiovascular coexisting conditions.[121] In this trial, there were similar rates of adverse cardiovascular events between groups, but higher all-cause mortality (hazard ratio [HR] 1·22 [95% CI 1·01–1·47]; p=0·04) and higher cardiovascular mortality (HR 1·34 [95% CI 1·03–1·73]; p=0·03) in the febuxostat group. Gout outcomes, including serum urate lowering and gout flares, were similar between groups. The results of this trial resulted in a black box warning by the FDA, in 2019, that health-care professionals should reserve febuxostat for patients who did not respond to or do not tolerate allopurinol, counsel patients about the cardiovascular risk with febuxostat, and advise them to seek medical attention immediately if they experience the symptoms of a cardiovascular event. The mechanisms of increased mortality and cardiovascular mortality in the febuxostat group remain unclear, particularly because approximately 85% of the deaths occurred when participants were not taking study medication.[121] In 2020, the European Medicines Agency-mandated trial of cardiovascular outcomes (FAST) was published; this trial of 6128 patients with gout, aged 60 years or older and with at least one additional cardiovascular risk factor, showed no increased risk of adverse cardiovascular events, all-cause mortality, or cardiovascular mortality with febuxostat compared with allopurinol.[130]

Uricosuric medications such as probenecid, sulfinpyrazone, and benzbromarone can be used as monotherapy or in combination with a xanthine oxidase inhibitor.[131,132] Uricosurics act by promoting urate excretion at the renal tubules, and urolithiasis can occur as a side-effect. Uricosurics should be avoided in patients with previous or current urolithiasis, and all patients taking uricosurics should maintain a high fluid intake (typically 2 L per day). Benzbromarone can, rarely, cause hepatotoxicity, and liver function test monitoring is required when prescribing this agent. Lesinurad, a uricosuric agent that specifically inhibits URAT1, is no longer marketed in the USA or Europe (marketing authorisation withdrawn at the request of the marketing-authorisation holder).

Pegloticase is a pegylated recombinant uricase administered as a fortnightly intravenous infusion. Pegloticase metabolises urate to soluble allantoin, leading to undetectable plasma urate concentrations in responders. Clinical trials of pegloticase showed improvement in gout flares, tophus size, activity limitation, and health-related quality of life over 6 months.[133] In the phase 3 clinical trials, 26% of patients receiving fortnightly intravenous infusions had infusion reactions, which were associated with anti-polyethylene glycol antibodies and with the loss of serum urate-lowering efficacy. For patients receiving pegloticase, serum urate should be measured before each infusion and treatment discontinued if concentrations increase to more than 6 mg/dL (0·36 mmol/L), particularly on two consecutive measurements. Clinical trials are currently underway to determine if concomitant immunosuppression with methotrexate, azathioprine, or mycophenolate can reduce the development of anti-drug antibodies and infusion reactions.[134]

Primary prevention of incident gout

Primary prevention of gout should be considered for those with asymptomatic hyperuricaemia, the most important risk factor for the development of gout. Medications indicated for related cardiometabolic conditions such as losartan, fenofibrate, and SGLT2 inhibitors have modest urate-lowering effects and also reduce incident gout.[135–137] Similarly, substantial weight loss (particularly in the context of bariatric surgery[138]) and the Dietary Approaches to Stop Hypertension diet reduce serum urate concentrations and, consequently, the risk of developing gout.[139] Incident gout can be prevented with urate-lowering therapy, such as febuxostat.[140] In addition, anti-inflammatory therapy, such as the IL-1β inhibitor canakinumab, can prevent incident gout without influencing serum urate concentrations.[141] However, urate-lowering or anti-inflammatory medications for the primary purpose of preventing incident gout are not currently recommended.[103]

Outcomes

Despite the high population burden of gout, treatment of this disease remains suboptimal worldwide.[24,26,29,142,143] Many patients have recurrent gout flares and do not receive regular urate-lowering therapy, which leads to poor health-related quality of life.[143] In the USA, only a third of people with gout are prescribed urate-lowering therapy.[26] In Australia, a concentration of serum urate under 0·36 mmol/L was documented in 22·4% of all people with gout over a 5 year period.[142] Even when urate-lowering therapy is prescribed, most UK patients have medication gaps.[144] The low adherence to urate-lowering therapy is likely to be due to many factors, including systems barriers (eg, time constraints, cost, and scarcity of incentives), the intermittent flaring nature of the illness, which can have long asymptomatic intercritical periods, and beliefs and incomplete knowledge about the illness and its management.[145]

Poorly controlled gout incurs substantial societal costs, with direct annual costs for those with frequent flares estimated to be US$26 890, and indirect costs of $13 164.[146] In addition, gout is associated with an increased risk of death, even after adjusting for age, sex, and comorbidities.[22] Cardiovascular disease is the major cause of increased mortality in patients with gout.[147] In contrast with rheumatoid arthritis, the mortality gap for gout has not improved over the past 20 years.[22]

Controversies and uncertainties

Although the underlying cause of gout (monosodium urate crystal deposition) is well documented and effective therapies are available, there are various uncertainties. For centuries, the dominant narrative about the cause of gout and its treatment has focused on dietary management.[148] However, some research suggests that diet plays only a small role in regulation of serum urate in the healthy population;[38] it is unknown whether these findings are also true for gout. Furthermore, randomised controlled trials of dietary interventions have not shown efficacy in people with gout.[117,149,150] Dietary management of gout is a research priority for patients,[151] and a current uncertainty.

A further controversy is whether high serum urate concentrations or gout play a direct role in the pathogenesis of associated medical conditions, such as cardiovascular disease, hypertension, or chronic kidney disease. Although many prospective observational studies have shown an association between hyperuricaemia or gout and development of many medical conditions, a causal role has not been established.[152] In addition, although low serum urate concentrations have been associated with neurodegenerative conditions, such as Parkinson's disease and Alzheimer's disease, in observational studies,[152] the relationship between gout and neurodegenerative diseases is currently unclear.[153,154]

A major advance in gout management has been the recognition that a treat-to-target approach with nurse-led care leads to impressive clinical outcomes for people with gout.[106] A UK clinical trial with nurse-led care that included an individualised education package, serum urate testing, and dose titration of urate-lowering therapy using a treat-to-target approach led to high uptake and adherence of urate-lowering therapy; after 2 years, 96% of participants in the nurse-led care group were taking urate-lowering therapy, compared with 56% in the usual care group.[106] Clinical outcomes including gout flares, tophi, and physical function were also better in the nurse-led care group. Nurse-led care was cost-effective, and potentially cost-saving after 5 years. A key ongoing uncertainty is how to embed effective strategies such as nurse-led care into clinical care. Different solutions might be needed for health-care systems across the world, and it can include pharmacists, Indigenous community health workers, or a multi-disciplinary team.[155–157]

Conclusion

Gout is a common and treatable rheumatic disease, caused by deposition of monosodium urate crystals in people with hyperuricaemia. Although presenting as an intermittent flaring condition, gout is a chronic disease. For patients with recurrent flares or tophaceous gout, long-term urate-lowering therapy with medications such as allopurinol leads to dissolution of monosodium urate crystals, ultimately resulting in prevention of gout flares and improved quality of life. Despite the availability of inexpensive and effective therapies, there are low rates of urate-lowering therapy initiation and persistence. Strategies such as nurse-led care are effective in delivering high-quality gout care and lead to major improvements in patient outcomes.

Contributors

ND completed the search and coordinated the manuscript writing. All authors wrote the manuscript and approved the final draft.

Declaration of interests

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